Table 2a. Timor-Leste: Public Sector Debt Sustainability Framework, Baseline Scenario, 2014-2037

(In percent of GDP, unless otherwise indicated)

	Actual					Projections									
	2014	2015	2016	Average 5/	Standard Deviation 5/	2017	2018	2019	2020	2021	2022	2017-22 Average	2027	2037	2023-37 Average
Public sector debt 1/	0.5	1.5	2.8			3.1	10.6	17.8	24.3	27.6	28.3		25.3	21.4	
of which: foreign-currency denominated	0.5	1.5	2.8			3.1	10.6	17.8	24.3	27.6	28.3		25.3	21.4	
Change in public sector debt	0.4	0.9	1.3			0.4	7.4	7.2	6.5	3.3	0.7		-0.9	-0.6	
Identified debt-creating flows	20.0	24.3	37.1			25.7	46.1	50.1	41.2	35.9	29.5		19.3	7.6	
Primary deficit	20.0	24.2	36.9	15.6	9.1	25.7	46.2	50.1	41.5	36.8	31.0	38.5	21.5	9.4	17.6
Revenue and grants	19.8	26.1	27.5			24.2	24.4	23.4	22.0	20.5	19.3		15.3	16.9	
of which: grants	6.7	7.2	6.1			5.7	5.6	5.6	5.6	5.6	5.6		5.6	4.8	
Primary (noninterest) expenditure	39.8	50.2	64.5			49.9	70.6	73.5	63.4	57.3	50.3		36.8	26.3	
Automatic debt dynamics	0.0	0.2	0.2			0.0	0.0	0.0	-0.3	-0.8	-1.5		-2.2	-1.8	
Contribution from interest rate/growth differential	0.0	-0.1	0.1			0.2	0.1	0.6	0.8	0.7	0.8		-1.2	-1.0	
of which: contribution from average real interest rate	0.0	0.0	0.0			0.0	0.0	0.0	0.1	0.1	0.0		0.1	0.1	
of which: contribution from real GDP growth	0.0	-0.1	0.1			0.2	0.0	0.5	0.7	0.6	0.7		-1.3	-1.1	
Contribution from real exchange rate depreciation	0.0	0.3	0.1			-0.2	-0.1	-0.6	-1.1	-1.6	-2.3		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-19.6	-23.4	-35.8			-25.4	-38.7	-42.9	-34.7	-32.6	-28.8		-20.2	-8.1	
Other Sustainability Indicators															
PV of public sector debt	...	...	1.4			1.6	6.7	11.7	16.2	18.7	19.5		18.2	15.4	
of which: foreign-currency denominated	...	...	1.4			1.6	6.7	11.7	16.2	18.7	19.5		18.2	15.4	
of which: external	...	...	1.4			1.6	6.7	11.7	16.2	18.7	19.5		18.2	15.4	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	20.0	24.2	36.9			25.8	46.3	50.5	42.0	37.4	31.7		22.6	10.5	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	4.9			6.7	27.4	49.9	73.9	91.4	101.1		119.0	91.2	
PV of public sector debt-to-revenue ratio (in percent)	...	...	6.3			8.7	35.5	65.5	99.0	125.4	142.1		187.2	127.2	
of which: external 3/	...	...	6.3			8.7	35.5	65.5	99.0	125.4	142.1		187.2	127.2	
Debt service-to-revenue and grants ratio (in percent) 4/	0.0	0.0	0.0			0.4	0.8	1.5	2.2	2.9	3.3		7.1	6.4	
Debt service-to-revenue ratio (in percent) 4/	0.0	0.0	0.0			0.5	1.0	1.9	3.0	4.0	4.6		11.1	8.9	
Primary deficit that stabilizes the debt-to-GDP ratio	19.5	23.2	35.6			25.4	38.7	42.9	35.0	33.5	30.3		22.4	10.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	-26.0	20.9	-7.9	0.4	13.5	-8.0	-0.7	-4.7	-3.7	-2.6	-2.6	-3.7	5.2	5.2	5.2
Average nominal interest rate on forex debt (in percent)	0.0	0.0	0.0	0.0	0.0	0.2	3.1	2.5	2.4	2.3	2.2	2.1	2.2	2.3	2.3
Average real interest rate on domestic debt (in percent)	...	...	...	0.0	0.0	...	...	...	...	...	...	0.0	...	...	0.0
Real exchange rate depreciation (in percent, + indicates depreciation)	5.1	59.2	4.8	3.2	25.4	-6.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	-3.2	-36.5	-3.3	3.3	22.6	8.8	4.9	8.0	8.4	8.7	11.1	8.3	6.0	6.0	6.0
Growth of real primary spending (deflated by GDP deflator, in percent)	23.7	52.7	18.1	9.6	17.5	-28.8	40.3	-0.7	-16.9	-12.1	-14.4	-5.4	1.1	2.4	0.8
Grant element of new external borrowing (in percent)	...	...	...	...	...	33.0	31.4	31.4	31.4	31.5	31.6	31.7	31.6	31.6	...

Sources: Country authorities; and staff estimates and projections.

1/ Covers gross debt of central government.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b. Timor-Leste: Sensitivity Analysis for Key Indicators of Public Debt, 2017-2037

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	2	7	12	16	19	20	18	15
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	2	-14	-30	-40	-47	-50	-63	-56
A2. Primary balance is unchanged from 2017	2	-7	-19	-25	-30	-31	-10	64
A3. Permanently lower GDP growth 1/	2	7	14	20	25	28	38	73
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	2	9	20	28	34	37	40	41
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	2	-8	-21	-16	-13	-11	-3	6
B3. Combination of B1-B2 using one half standard deviation shocks	2	-13	-31	-25	-21	-18	-5	11
B4. One-time 30 percent real depreciation in 2018	2	6	10	14	16	17	17	14
B5. 10 percent of GDP increase in other debt-creating flows in 2018	2	14	19	23	26	26	23	17
PV of Debt-to-Revenue Ratio 2/								
Baseline	7	27	50	74	91	101	119	91
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	7	-58	-132	-186	-237	-270	-402	-277
A2. Primary balance is unchanged from 2017	7	-30	-83	-116	-146	-160	-65	380
A3. Permanently lower GDP growth 1/	7	30	57	89	117	138	221	353
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	7	38	80	121	154	178	238	225
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	7	-33	-88	-73	-63	-56	-19	38
B3. Combination of B1-B2 using one half standard deviation shocks	7	-51	-130	-112	-99	-89	-29	62
B4. One-time 30 percent real depreciation in 2018	7	26	45	65	80	90	109	84
B5. 10 percent of GDP increase in other debt-creating flows in 2018	7	56	80	105	125	135	148	103
Debt Service-to-Revenue Ratio 2/								
Baseline	0	1	1	2	3	3	7	6
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	0	1	-1	-4	-5	-7	-17	-20
A2. Primary balance is unchanged from 2017	0	1	0	-2	-3	-4	-5	11
A3. Permanently lower GDP growth 1/	0	1	2	3	4	4	11	18
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	0	1	2	3	4	5	12	13
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	0	1	-1	-2	-2	-1	-4	2
B3. Combination of B1-B2 using one half standard deviation shocks	0	1	-1	-4	-3	-3	-6	2
B4. One-time 30 percent real depreciation in 2018	0	1	2	3	4	5	10	10
B5. 10 percent of GDP increase in other debt-creating flows in 2018	0	1	2	3	4	4	9	7

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined exclusive of grants.